Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/02/2024
PI Name:	Davis, Anthony Ph.D.		
Project Title:	Evaluation of the Cellular and Molecular Responses of Lung and Liver Cells Derived From iPSCs With Varying Intrinsic Radiosensitivity to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHCSR, Omnibus2. Human Health Countermeasures and Space Radiation Topics Appendix C; OMNIBUS2-Appendix D
Start Date:	06/01/2023	**End Date:**	12/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 10/22/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Jun Ph.D. (University of Texas Southwestern Medical Center at Dallas) Story, Michael Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	80NSSC23K1018		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Deleterious mutations in genes required for the DNA damage response (DDR) cause radiosensitivity and are associated with increased cancer risk. However, variability in intrinsic radioresponse across the human population is likely governed by other factors besides loss of function of DDR proteins, including genetics, epigenetics, changes in functional/metabolic pathways, and slight perturbations in the DDR. A key unanswered question is if a relationship between intrinsic radiosensitivity and risk for cancer from radiation exposures on Earth and in deep space exist. To address this, we will use a novel genetically diverse panel of primary human skin fibroblasts (PHSFs) developed from 90 individuals that have a four-fold difference in radioresponse to gamma rays. These PHSFs will be utilized to generate immortalized PHSFs and induced pluripotent stem cells (iPSCs) and iPSC-derived hepatocytes and lung epithelial cells whose intrinsic radiosensitivity is the same as that of the original PHSF from which it was derived. These two cell types were chosen based upon their risk for induction of space radiation-induced lethal cancers. All assays will be performed with low (gamma rays) and high (5 ion galactic cosmic ray (GCR) simulator and representative component ions) linear energy transfer (LET) radiations. A myriad of experimental procedures will be performed, including clonogenic survival, DNA repair, DDR signaling, genome stability, mutation frequency and cellular and oncogenic transformation assays, transcriptome, whole genome sequencing, DNA methylation, and RNAseq analysis. The goal is to identify consequential molecular or genetic features of intrinsic radioresponse that are associated with the risk and/or protection from space radiation-induced oncogenic transformation and carcinogenesis. The ability to identify individuals at greatest risk (or lowest risk) for carcinogenesis induced by low and high LET irradiations would be invaluable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Deleterious mutations in genes required for the DNA damage response (DDR) cause radiosensitivity and are associated with increased cancer risk. However, variability in intrinsic radioresponse across the human population is likely governed by other factors besides loss of function of DDR proteins, including genetics, epigenetics, metabolic pathways, and slight perturbations in the DDR. A key unanswered question is if a relationship between intrinsic radiosensitivity and risk for cancer from radiation exposures on earth and in deep space exist. To address this, we will use a panel of 90 primary human skin fibroblasts (PHSFs) that have a four-fold difference in radioresponse to gamma-rays to generate immortalized PHSFs and iPSCs and iPSC-derived hepatocytes and lung epithelial cells generated from the PHSFs. We will use this panel of PHSFs and derivatives from these parental cells (hTERT immortalized, iPSC, and iPSC-derived lung epithelial cells and hepatocytes) to mechanistically examine the role that intrinsic radiosensitivity plays in the cellular response to low (gamma-rays at UT Southwestern) mixed (5 ion Galactic Cosmic Ray simulation (GCRsim) and representative (component ions of the GCRsim at the NSRL) radiation. These experiments will allow us to characterize how inter-individual radioresponses contribute to the response to mixed and individual species of the space radiation environment and its impact on radiation-associated cancer risk, as monitored by genomic instability, which is a hallmark of cancer. The ultimate goal is to identify candidate biomarkers to determine individual radiation susceptibility/resistance to radiation, in particular space radiation. We believe this will result in initial recommendations for clinical practice guidelines incorporating biomarkers into standard measures for future space missions.
Task Progress:	The response to ionizing radiation (IR) varies among individuals. This is supported by studies showing that individuals with deleterious mutations in genes required for the DNA damage response (DDR) cause radiosensitivity and these defects are associated with increased cancer risk. However, variability in intrinsic radioresponse across the human population is observed in individuals without gross defects in DDR genes, which suggests that other factors besides loss of function of DDR proteins govern the radioresponse, including genetics, epigenetics, metabolic pathways, and slight perturbations in the DDR. A key unanswered question is does a relationship between intrinsic radiosensitivity and risk for cancer from radiation exposures on Earth and in deep space exist? Conversely, are those individuals that show increased radioresistance, less prone to radiation-induced carcinogenesis? To address these questions, we will use a panel of 90+ primary human skin fibroblasts (PHSFs) that have a four-fold difference in radioresponse to ?-rays to generate immortalized PHSFs and induced pluripotent stem cells (iPSCs) and iPSC-derived hepatocytes and lung epithelial cells generated from the PHSFs. A significant power of our panel of PHSFs is that it contains cells from different sexes, ethnic groups, and a range of chronological ages (from childhood to late adulthood), and thus it will allow us to explore heterogeneity in the cellular response to radiation. This novel genetically diverse panel of PHSFs will allow us to determine how radiosensitivity impacts the risk for radiation-induced cancer, particularly radiations induced by heavy ions associated with deep space travel in two cell types representative of organs at risk for HZE-induced lethal cancers (lung and liver). Further, we believe this panel of cell lines is well suited to examine questions concerning radioresponse and DNA damage-induced genomic instability and carcinogenesis by low and high LET irradiation. We believe that by understanding the drivers of intrinsic radioresponse, identified features may serve as biomarkers of oncogenic risk for both low and high linear energy transfer (LET) radiation exposures. The ultimate goal is to identify candidate biomarkers to determine individual radiation susceptibility/resistance to radiation, in particular space radiation. We believe this will result in initial recommendations for clinical practice guidelines incorporating biomarkers into standard measures for future space missions. The primary objectives of the Definition Phase of this grant are: (1) Generate iPSCs from the ten selected PHSFs (three radiosensitive, three radionormal, three radioresistant, and one A-T (positive control for DDR defective); (2) Assess the radioresponse of the 10 selected lines and verify the phenotype is conserved following iPSC generation; and (3) Determine if the IR-induced DNA damage response is conserved in the ten selected lines following iPSC generation. During this funding cycle, we have generated and verified eight total iPSCs for the project and continue to generate more iPSCs. Importantly, we have verified that in general, the iPSCs have retained their parental radioresponse following reprogramming of the fibroblasts to iPSCs. Lastly, we have characterized DDR signaling in these eight PHSFs and iPSCs and we found that the DDR signaling is conserved in the iPSCs from the parental lines. In conclusion, we are on track to complete the Definition Phase of this grant.
Bibliography Type:	Description: (Last Updated: 04/30/2025)
Abstracts for Journals and Proceedings	Wang S-Y, Lu H, Schmitz D, Wu J, Story MD, Davis AJ. "Evaluation of the cellular and molecular responses of iPSCs derived from primary human skin fibroblast with varying intrinsic radioresponse to radiation." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024